

Integration — Lesson 34 Test: Volumes I — Disks & Washers

10 questions · show your setup and work in the box · exact answers (fractions and π), no decimals · square the radius FIRST, then integrate

Name: _____ Date: _____ Score: _____ / 10

1. The region under $y = x^2$ from $x = 0$ to $x = 2$ is revolved about the x-axis. Find the volume. 1 pt

2. The region under $y = \sqrt{2x}$ from $x = 0$ to $x = 3$ is revolved about the x-axis. Find the volume. 1 pt

3. Revolve the constant $y = 5$ on $[2, 6]$ about the x-axis. Find the volume with the disk formula, then confirm it against geometry's cylinder formula $V = \pi r^2 h$. State the radius and the height you used. 1 pt

4. Revolve the region under $y = x$ on $[0, 4]$ about the x -axis. Find the volume with calculus, ^{1 pt} then check it against geometry's cone formula $V = (1/3)\pi r^2 h$.

5. A loaf-shaped solid sits along the x -axis from $x = 0$ to $x = 6$, with x measured in ^{1 pt} centimeters. Its cross-section area at position x is $A(x) = 4 + x$ square centimeters. (a) Use $V = \int A(x) dx$ to find the volume. (b) Explain in one sentence why the answer comes out in cubic centimeters.

6. The region between $y = x + 1$ (top) and $y = 1$ (bottom) on $[0, 2]$ is revolved about the ^{1 pt} x -axis. Find the volume. (Expand the squared radius carefully.)

7. The region between $y = 3x$ and $y = x$ on $[0, 1]$ is revolved about the x-axis. A student writes $V = \pi \int_0^1 (3x - x)^2 dx$. (a) Compute the student's value. (b) Compute the correct value. (c) Using $R = 3$ and $r = 1$ as numbers, show in one line why $R^2 - r^2$ and $(R - r)^2$ are not the same thing. 1 pt

8. The region enclosed by $y = \sqrt{x}$ and $y = x^2$ is revolved about the x-axis. No limits are given: first find where the two curves intersect, say which one is the outer radius, then find the volume. 1 pt

9. Write — but do NOT evaluate — an integral for the volume when the region under $y = \sin x$ from $x = 0$ to $x = \pi$ is revolved about the x-axis. Name the method you used and state the radius. 1 pt

10. The region between $y = \sqrt{25 - x^2}$ (the top half of a circle of radius 5) and the line $y = 3$ is revolved about the x-axis, drilling a round hole straight through a sphere. (a) Find the x-values where the curve meets the line. (b) Find the volume of the resulting ring. 1 pt
